REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Reflection of Light Practice Questions

Grade 10 Advanced | Module 16 Lesson 1 | Reflection of Light

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: θi = θr, measured from the normal. Plane mirror: di = −do; hi = ho; magnification = +1.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 3–8.

P01. The angle of incidence is measured between the incident ray and the:

A. Mirror surface B. Normal
C. Image plane D. Object height

P02. For specular reflection, the surface is smooth compared with the:

A. Distance to the Sun B. Observer's height
C. Wavelength of light D. Mirror frame

P03. An image formed by a plane mirror is:

A. Real and inverted B. Real and enlarged
C. Virtual and inverted D. Virtual and upright

P04. Diffuse reflection occurs because:

A. Local surface normals differ across a rough surface B. The law of reflection fails
C. Light loses all wave properties D. Every ray is absorbed

P05. A plane-mirror image is located:

A. At the mirror surface always B. As far behind the mirror as the object is in front
C. Twice the object distance in front D. At a fixed focal point

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P06. A light ray has incidence angle 20° to the normal at a plane mirror. Find the reflection angle to the normal.

A. 40 ° B. 10 °
C. 20 ° D. -20 °

P07. A ray makes 30° with the mirror surface. Find its incidence angle measured from the normal.

A. 120 ° B. 30 °
C. -60 ° D. 60 °

P08. A ray strikes a plane mirror at incidence angle 20°. Find the smaller angle between the incident and reflected ray arms at the surface.

A. 40 ° B. 80 °
C. 20 ° D. -40 °

P09. An object is 24 cm in front of a plane mirror. Find the distance between the object and its image.

A. 96 cm B. 48 cm
C. 24 cm D. -48 cm

P10. An object moves 6 cm closer to a stationary plane mirror. By how much does its separation from the image decrease?

A. 24 cm B. 6 cm
C. 12 cm D. -12 cm

P11. State the law of reflection.

P12. Distinguish specular reflection from diffuse reflection.

P13. Why is a plane-mirror image virtual?

P14. State the size and orientation of a plane-mirror image.

P15. Why can a rough page be seen from many directions?

P16. A light ray has incidence angle 25° to the normal at a plane mirror. Find the reflection angle to the normal.

P17. A light ray has incidence angle 30° to the normal at a plane mirror. Find the reflection angle to the normal.

P18. A ray makes 35° with the mirror surface. Find its incidence angle measured from the normal.

P19. A ray makes 40° with the mirror surface. Find its incidence angle measured from the normal.

P20. A ray strikes a plane mirror at incidence angle 25°. Find the smaller angle between the incident and reflected ray arms at the surface.

P21. A ray strikes a plane mirror at incidence angle 30°. Find the smaller angle between the incident and reflected ray arms at the surface.

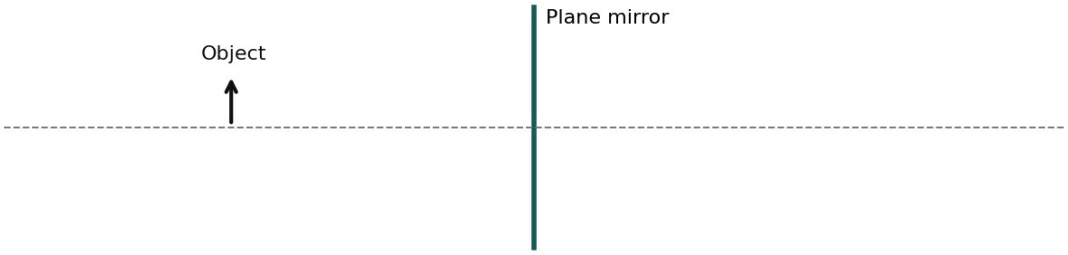
P22. An object is 36 cm in front of a plane mirror. Find the distance between the object and its image.

P23. An object is 48 cm in front of a plane mirror. Find the distance between the object and its image.

P24. An object moves 9 cm closer to a stationary plane mirror. By how much does its separation from the image decrease?

P25. An object moves 12 cm closer to a stationary plane mirror. By how much does its separation from the image decrease?

P26. Draw a plane mirror, an object point and two rays to locate its image. Label the normal for one ray.



P27. A student measures 35° between a ray and the mirror surface and calls it the incidence angle. Correct the value.

P28. Does diffuse reflection violate the law of reflection? Explain.

P29. Why does moving a plane-mirror object closer not make its image physically larger?

P30. Design a simple experiment to test the law of reflection.